

“WLAN Testing and Associated Challenges”

Ashwin Gopinath of EFY asked industry experts four questions on WLAN testing. This special interview clubs their responses and gives you the big picture. Here is what Abhay Samant of NI, Madhukar Tripathi of Anritsu, J.K. Baldua of Scientech, Naresh Narasimhan of Tektronix, and Naresh Kumar of Agilent have to say about the challenges, trends and effects of the latest 802.11ac standards on WLAN testing



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What are some of the unique challenges in this field making it different from the other areas of T&M?

Samant: Like previous standards, the newest 802.11ac standard builds on the strategies of higher-order modulation types, wider bandwidth and more spatial streams to increase data rates over existing 802.11n products. All of these evolving challenges imply that design and test engineers will need to design their unique measurements and not rely on vendor-defined instruments.

Tripathi: If the user has developed something for which over-the-air communication is the major application and has deployed the system in field, it becomes very important to verify the system performance over the air. Hence it becomes utmost important for the user to actually test the deployed system when it is in field.

Baldia: One key challenge is detecting low-level signals buried in plenty of noise along with having to segregate each frequency carrying the signal separately in a bunch of frequency spectrum. It is a challenge as this needs a much higher bandwidth resolution as well as the ability to distinguish the very closely spaced frequencies.

Narasimhan: The unique challenges in the RF/wireless domain are dealing with the transient RF behaviour, the innate ability to measure very low-power signals, dealing with a host of interference sources that induce noise and other non-idealities, and certain issues that affect other fac-

tors as well termed parasitic issues like parasitic oscillation and noise mitigation.

Kumar: Wireless domain is an evolving field as the standards are being updated on a real-time basis. This introduces challenges in T&M, as this industry needs to be ahead of the curve in providing test solutions for evolving technologies.

This means that we have to work very closely with customers and industry bodies to align our product development strategy to the dynamic testing needs as they evolve.

What are the major parameters to be tested in RF/wireless?

Samant: For WLAN testing, two types of parameters are to be tested: One on the transmit side and the other on the receive side. Test parameters on the transmission side are adjacent channel power ratio, transmit error vector magnitude, spectral emission mask, group delay, etc. On the receiver side, parameters tested relate to the integrity of the signal once it reaches the receiver side. These include sensitivity, dynamic frequency selection, etc.

Tripathi: The application is the key factor. For wireless, some of the key parameters depending upon the applications can be field strength, occupied bandwidth, power density, accuracy of frequency and amplitude of the signal. Further, if the user has applications dedicated to a particular